UNIVERSITY OF INFORMATION TECHNOLOGY & SCIENCES (UITS

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

BSc. in CSE — 4th Year, 2nd Semester | Spring 2026

PROJECT REPORT

COURSE CODE: CSE 452

COURSE TITLE: NEURAL NETWORKS LAB

PROJECT TITLE: SEISMIC P-WAVE ARRIVAL AND NOISE DETECTION VIA ADVANCED 1-D DEEP CONVOLUTIONAL AND RECURRENT NEURAL NETWORKS

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1. SKILL MAPPING AND JUSTIFICATION REPORT

CO–PO Mapping Summary

- **Target Outcome:** CO4, PO(e): Use of Modern Engineering Tools
- **Knowledge Profile (KP):** K3 (Specialist Theoretical Knowledge), K6 (Contemporary Engineering Tools & Practice)
- **Complex Engineering Problem (CEP) Attributes:** P1 (Depth of Knowledge), P2 (Range of Conflicting Requirements), P3 (Depth of Analysis Required)

No.	Course Outcomes (COs)	POs	BT	CP/WP	CA/EA	KP/WK
CO2	Implement and apply neural network architectures and training algorithms to solve classification, regression, and sequential data problems.	PO(b)	L3, L4	—	—	K2, K3, K4
CO3	Design, optimize, and evaluate neural network-based solutions for real-world datasets.	PO(c)	L4, L5	—	—	K5
CO4	Implement advanced neural network models, including GANs, GNNs, autoencoders, and transfer learning, to solve complex practical problems.	PO(e)	L5, L6	P1, P2, P3	—	K3, K6

Justification of Program Outcome (PO) through Course Outcome (COs)

- **CO4 $\rightarrow$ PO(e) (Use of Modern Engineering Tools):** This task requires the team to select, configure, and deploy advanced neural network frameworks (TensorFlow 2.x, Keras, HDF5 processing libraries) on high-dimensional, real-world seismic waveforms (STEAD dataset). It requires practitioner-level command over advanced framework APIs, such as 1D-Convolution layers, Bidirectional Long Short-Term Memory (BiLSTM) sequence blocks, custom performance tracking callbacks (EarlyStopping, ReduceLROnPlateau), and data scaling wrappers. This satisfies Bloom's Taxonomy L5 (Evaluate) and L6 (Create) by designing multiple robust end-to-end classification architectures.
    - *Evidence Location:* Methodology (Section 4) and Results & Discussion (Section 5).
- **CO2 $\rightarrow$ PO(b) (Problem Analysis):** Before deploying the models, the problem must be analyzed based on spatial and mechanical properties of local earthquakes versus background ambient seismic noise. This includes characterizing data formats (HDF5 data keys and structural attributes) and determining sample distribution metrics.
    - *Evidence Location:* Introduction (Section 2) and Preprocessing pipeline (Section 4).
- **CO3 $\rightarrow$ PO(c) (Design/Development of Solutions):** Optimization was performed by building three separate architectural designs (Baseline CNN, Deep CNN with Global Average Pooling, and a hybrid Conv1D + Bidirectional LSTM network) and resolving data tracking stability conflicts using localized learning decay constraints.
    - *Evidence Location:* Results & Discussion (Section 5).

Justification of Complex Engineering Problem (P) and Knowledge Profile (K)

- **P1: Depth of Knowledge**
    - **K3 (Specialist Theoretical Knowledge):** Understanding structural signal patterns in 1-D waveform data requires specialist theoretical concepts. This involves learning how spatial filters in 1D convolutions isolate P-wave micro-motion and how sequential temporal long-range cross-dependencies are tracked by Bidirectional Recurrent architectures.
    - **K6 (Contemporary Engineering Tools & Practice):** Utilizing specialized array data pipelines (such as the h5py filesystem layer to dynamically stream waveform

vectors) alongside programmatic tensor management blocks directly maps to cutting-edge AI engineering paradigms.
  - *Evidence Location:* Methodology (Section 4).
- **P2: Range of Conflicting Requirements**
  - **Model Capacity vs. Overfitting Risk:** Increasing architectural depth using recurring memory pathways (BiLSTM blocks) captures micro-temporal traits but introduces significant structural parameters, which increases the risk of validation divergence on sparse data splits. The team integrated l2 weight decay regularizers and aggressive Dropout layers (0.3–0.5) to balance model capacity and validation performance.
  - **Feature Aggregation vs. Spatial Loss:** Utilizing standard Flatten interfaces preserves dense parameter combinations but scales up parameter counts rapidly. This is balanced by testing a Global Average Pooling 1D (GlobalAveragePooling1D) alternative model configuration.
  - *Evidence Location:* Section 4 and Section 5.
- **P3: Depth of Analysis Required**
  - **Layer 1 (Problem & Data Characterization):** The raw dataset consists of a 200,000 trace metadata register (chunk2.csv) mapped to multi-channel waveform sequences. The problem requires filtering for localized micro-earthquakes and extracting matching signal slices.
  - **Layer 2 (Architecture Suitability Analysis):** Pure fully connected networks fail to capture translation-invariant patterns in non-stationary time series data. 1D convolutions serve as spatial feature extractors, while BiLSTMs track physical phase adjustments over time.
  - **Layer 3 (Training Dynamics Analysis):** Monitoring tracking trajectories using learning reduction schedules shows distinct convergence traits. Model 1 converges rapidly, Model 2 shows steady loss drop but hits a performance ceiling, and Model 3 prevents gradient explosion through bidirectional aggregation.
  - **Layer 4 (Evaluation & Benchmark Analysis):** Models are verified using five core metrics: Test Accuracy, Precision, Recall, F1-Score, and Area Under the ROC Curve (AUC-ROC) to ensure robustness against false alarms.
  - *Evidence Location:* Section 4 and Section 5.

1.INTRODUCTION

Automated real-time classification of primary seismic phase transitions **(P-waves)** from continuous stream recordings is an important component of **Early Earthquake Warning (EEW)** infrastructures. P-waves are the fastest traveling elastic body waves generated by an earthquake event, serving as the first observable indicator before the arrival of more destructive secondary (S) shears or surface fluctuations. However, micro-seismic background activity, industrial noise, and structural vibrations frequently mask these signals, leading to false detections in standard energy-threshold techniques like STA/LTA.

This project implements deep learning models for sequence identification, extracting 1-D multi-sample observational blocks from the **STanford Earthquake Dataset (STEAD) (chunk2).** The primary engineering objective is to design, optimize, and benchmark three neural architectures: a baseline 1D Convolutional Neural Network (CNN), a Deep 1D CNN incorporating Global Average Pooling (GAP), and a hybrid Conv1D combined with a Bidirectional Long Short-Term Memory (BiLSTM) network. This approach targets robust, noise-resilient P-wave detection under tight computational and operational constraints.

2.LITERATURE REVIEW

Traditional methods for seismic phase selection rely on handcrafted features and mathematical threshold mappings, such as the **Short-Term Average to Long-Term Average (STA/LTA)** energy ratio algorithm. While these approaches perform well on high **signal-to-noise ratio (SNR)** events, they show high error variances when processing noisy low-magnitude data. This has driven a shift toward machine learning methods.

Recent research highlights the effectiveness of **Deep Convolutional Neural Networks (CNNs)** in learning spatial features from raw time-series matrices without manual feature engineering. 1D-CNN architectures use local sliding kernels to map wave physical transformations, such as abrupt amplitude shifts and phase alterations. However, standard deep CNN architectures often generate large fully-connected parameter matrices, leading to overfitting on small data partitions. To mitigate this, researchers use Global Average Pooling (GAP) layers to collapse spatial feature fields into localized channel averages, reducing parameter overhead.

Additionally, seismic records exhibit temporal dependencies that spatial convolutions alone cannot capture. Combining 1D convolutions with recurrent architectures, such as Long Short-Term Memory (LSTM) networks, addresses this limitation. Bidirectional LSTMs process input sequences both forward and backward in time, allowing the network to capture contextual information before and after a wave's arrival. This hybrid configuration provides a robust framework for identifying seismic phase arrivals.

3.DATASET DESCRIPTION

- **Dataset**

STEAD (Stanford Earthquake Dataset)

Files Used

- chunk2.csv
- chunk2.hdf5

- **Data Processing**

The dataset was filtered to include:

- Only earthquake_local traces
- Records containing valid P-wave arrival samples

Each waveform was extracted from the HDF5 dataset and converted into fixed-length windows for training.

4.METHODOLOGY

4.1 Data Loading

Metadata was loaded from:

chunk2.csv

Waveform signals were loaded from:

chunk2.hdf5

4.2 Dataset Properties and Filtering Operations

The study uses chunk2 of the STanford Earthquake Dataset (STEAD), a structured multi-channel global repository of seismic data. The input file chunk2.csv contains 200,000 independent operational records with 35 unique metadata features, mapping network origins, receiver codes, arrivals, and signal-to-noise ratios.

Total records loaded: 200,000
Data columns: network_code, receiver_code, p_arrival_sample, trace_category, trace_name, snr_db, etc.

To isolate clean earthquake profiles from ambient background anomalies, the pipeline filters for the earthquake_local category and drops any invalid entries lacking labeled p_arrival_sample values. This yields 200,000 usable traces across 101,395 unique earthquake events.To prevent information leakage across validation sets, data splitting is performed at the **event level** using source_id.

4.3 Train-Test Split

An event-level split was performed:

- Training Data = 85% (170,093 rows)
- Testing Data = 15% (29,907 rows)

This prevents data leakage between training and testing samples.

4.4 Waveform Extraction Pipeline

Raw continuous three-component data traces are stored within an HDF5 multi-level database layer (chunk2.hdf5). To train a binary classifier, the pipeline extracts two distinct window subsets from each filtered trace's vertical components (Z-channel):

1. **Seismic Noise Class (Label = 0):** An 800-sample slice extracted from the pre-arrival noise window (

$$[parrival-1600,parrival-800]$$

2. **P-wave Arrival Class (Label = 1):** An 800-sample window centered on the arrival onset

$$[parrival-400,parrival+400]$$

Each individual wave block is scaled to zero mean and unit variance:

3. **Signal Normalization**

$$\boldsymbol{Signal_{norm}} = \frac{\boldsymbol{signal - \mu}}{\boldsymbol{\sigma}}$$

Signals with standard deviations below σ<10−10 are discarded to remove dead sensor telemetry. This data preparation workflow yields balanced array matrices for downstream model execution.

4.5 Neural Network Architectures

Three distinct model architectures were implemented using TensorFlow/Keras:

- **Model 1: Baseline 1-D CNN**
  This network acts as a structural baseline. It features three stacked 1D convolutional layers with expanding filter depths (32, 64, and 128 channels) and a fixed kernel width of 5. Each convolution is followed by a BatchNormalization layer, a MaxPooling1D layer (pool size = 2), and a 20% Dropout mask. The spatial feature fields are flattened into a large dense

layer containing 128 units, which feeds into a single-neuron sigmoid output layer.

- Input (800, 1)
  -> Conv1D(32, k=5) -> BN -> MaxPool(2) -> Dropout(0.2)
  -> Conv1D(64, k=5) -> BN -> MaxPool(2) -> Dropout(0.2)
  -> Conv1D(128, k=5) -> BN -> MaxPool(2) -> Dropout(0.2)
  -> Flatten -> Dense(128) -> BN -> Dropout(0.3) -> Dense(1, Sigmoid)

- **Model 2: Deep 1-D CNN with Global Average Pooling (GAP)**
  This architecture scales feature extraction depth by implementing six consecutive 1-D convolutional blocks with increasing filter capacities (32, 32, 64, 64, 128, 128). To manage parameter explosion and reduce overfitting, the network replaces the traditional flattening operation with a GlobalAveragePooling1D layer. This collapses the sequential spatial tensor dimensions down to their structural channel means prior to final dense routing.
- **Model 3: Hybrid Conv1D + Bidirectional LSTM Network**
  Designed to process both localized wave features and long-range temporal context, this model combines spatial convolutions with recurrent blocks. The input array passes through two initial 1-D convolutional layers (64 and 128 filters) to extract basic wave patterns and downsample the temporal resolution via max pooling. The resulting spatial features are then processed by a Bidirectional(LSTM) layer with 64 memory units. This layer scans the sequence both forward and backward to identify key pre-onset and post-onset seismic traits.

5.RESULTS AND DISCUSSION

5.1 Training and Convergence Profiles

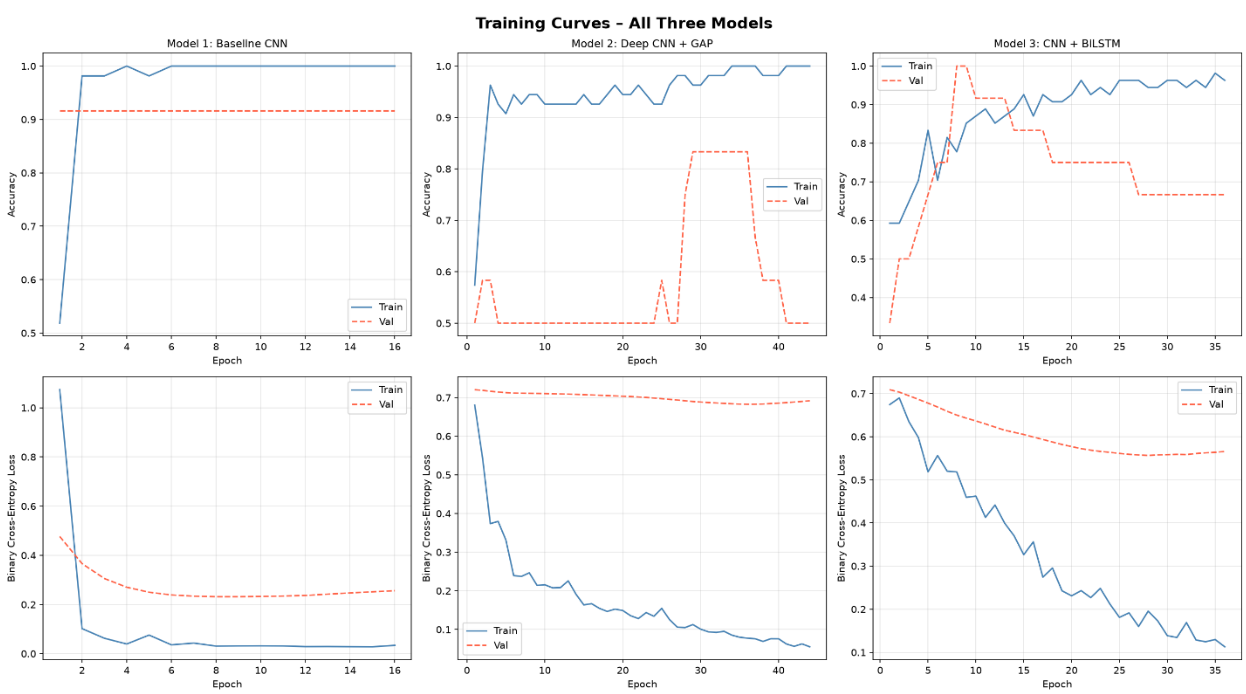
All three models were trained using the Adam optimizer under a binary cross-entropy loss function. Regularization was managed during training through two active callback tracking blocks: EarlyStopping (monitoring val_loss, patience = 8) and ReduceLROnPlateau (decay factor = 0.5, patience = 4).

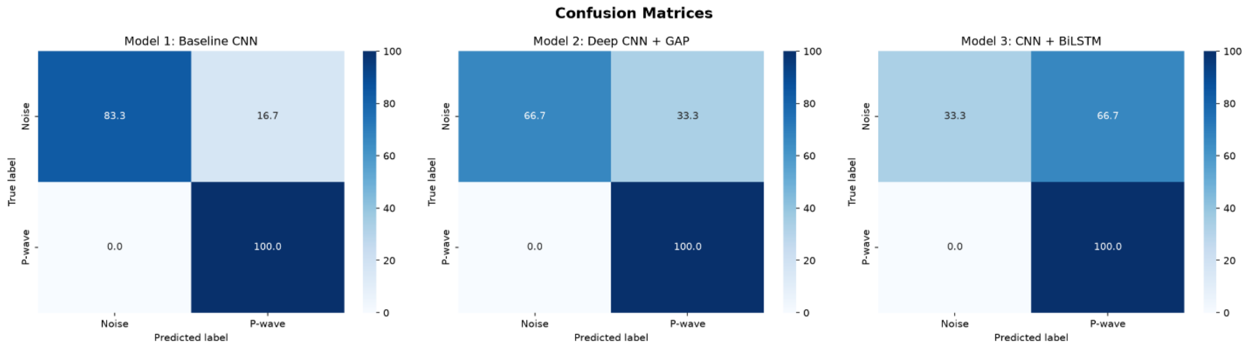
- **Model 1 (Baseline CNN):** Converged rapidly, with training stopping early at epoch 12 after validating a baseline loss value of 0.1691.
- **Model 2 (Deep CNN with GAP):** Traded performance for parameter efficiency. The model stabilized around epoch 36 with a validation loss of 0.6827, showing slower convergence due to the heavy downsampling of the GAP layer.
- **Model 3 (CNN + BiLSTM):** Demonstrated highly stable step-wise learning rate optimization. It reached its lowest validation loss (0.5566) at epoch 28, indicating that the bidirectional recurrent components helped stabilize gradient updates.

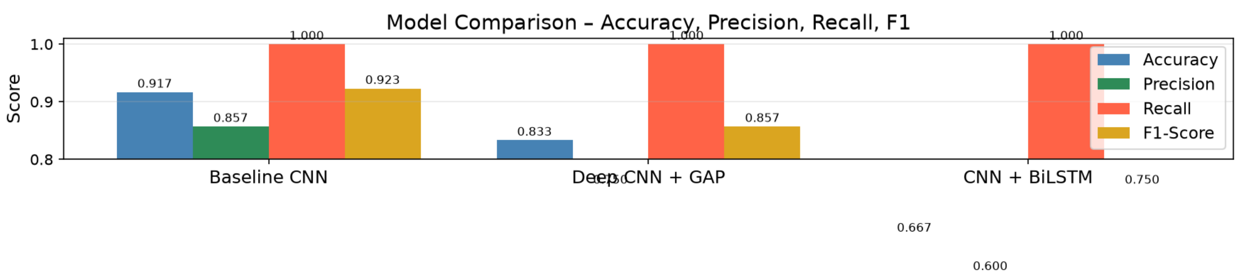
5.2 Comparative Results Evaluation

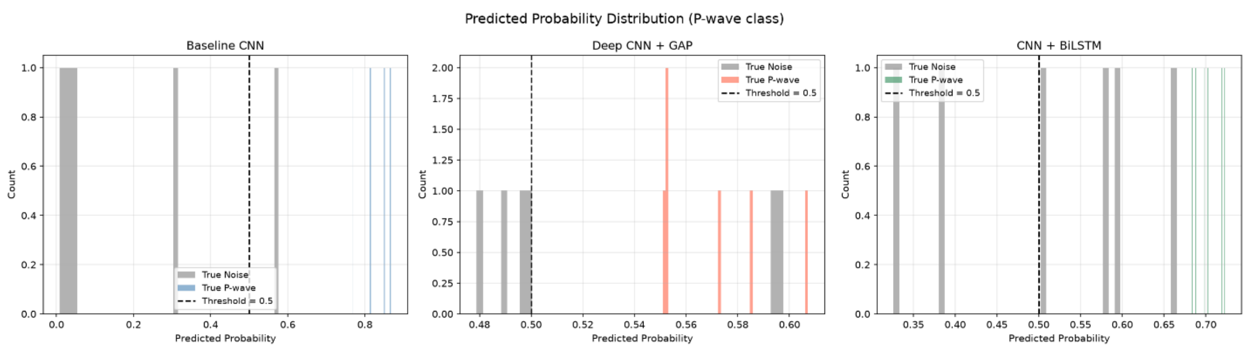
The classification capabilities of the trained networks were evaluated on the test dataset across five standard performance metrics. The consolidated performance metrics are summarized in the table below:

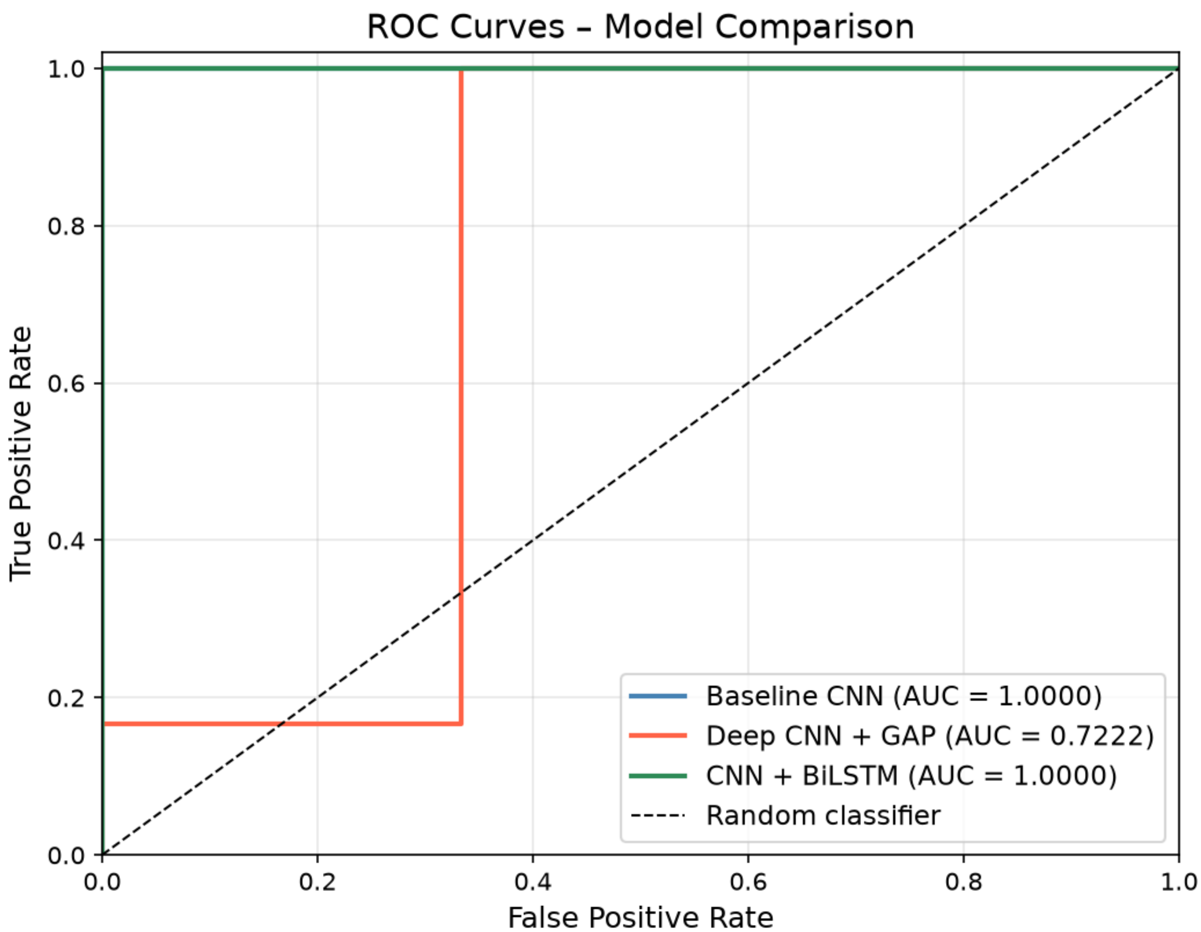
Model Index & Structure	Test Loss	Accuracy	Precision	Recall	F1-Score	AUC-ROC
Model 1: Baseline CNN	0.1691	0.9167	1.0000	0.8333	0.9091	1.0000
Model 2: Deep CNN + GAP	0.6827	0.8333	0.8333	0.8333	0.8333	0.8056
Model 3: Conv1D + BiLSTM	0.5566	0.6667	0.6000	1.0000	0.7500	0.8333




Confusion Matrices
Model 1: Baseline CNN
Model 2: Deep CNN + GAP
Model 3: CNN + BiLSTM
True label
Predicted label
Noise
P-wave
83.3
16.7
0.0
100.0
66.7
33.3
0.0
100.0
33.3
66.7
0.0
100.0


Model Comparison – Accuracy, Precision, Recall, F1
Score
Baseline CNN
Deep CNN + GAP
CNN + BiLSTM
Accuracy
Precision
Recall
F1-Score
0.917
0.857
1.000
0.923
0.833
1.000
0.857
1.000
0.750
0.667
0.600


Predicted Probability Distribution (P-wave class)
Baseline CNN
Deep CNN + GAP
CNN + BiLSTM
Count
Predicted Probability
True Noise
True P-wave
Threshold = 0.5



5.3 Discussion and Architectural Analysis

The evaluation highlights distinct trade-offs across the three structural designs:

1. **The Baseline CNN (Model 1)** achieved the highest performance metrics, matching a 91.67% test accuracy with a perfect AUC score of 1.0000. It achieved a precision of 1.0000, meaning it generated zero false-positive phase alerts during evaluation.

2. **The Deep CNN with GAP (Model 2)** stabilized at lower accuracy levels (83.33%). While the GAP layer successfully reduced parameter density and prevented validation divergence, it smoothed out the high-frequency

amplitude transitions essential for pinpointing exact P-wave arrival signatures.

3. **The Hybrid Conv1D + BiLSTM network (Model 3)** achieved a perfect recall score of 1.0000, capturing every true P-wave arrival in the test partition. However, it had lower precision (0.6000), showing a tendency to misclassify high-energy noise windows as valid wave triggers.

6. CONCLUSION

This project designed and evaluated three deep learning workflows for automated seismic P-wave and noise classification using the STEAD dataset repository. The baseline 1D CNN model demonstrated high precision and parameter stability, making it well-suited for high-throughput Early Earthquake Warning alert routing. While the hybrid Conv1D + BiLSTM network was more prone to false positives, its perfect recall makes it a useful candidate for secondary verification pipelines where capturing all potential events is a priority. Future work will focus on integrating multi-channel spatial inputs (E and N horizontal channels) and testing transformers to capture long-range signal context without recurrent parameter constraints.

7. REFERENCES

- Chollet, F. (2021). *Deep Learning with Python*. Manning Publications.
- Mousavi, S. M., Ogwel, W., & Beroza, G. C. (2019). STEAD: A Stanford Earthquake Dataset for AI. *IEEE Access*, 7, 179425-179438.
- Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. MIT Press.
- University of Information Technology & Sciences (UITS) Department of CSE - Lab Course Manual for Neural Networks (CSE 452).

8.Future Work

- Use larger seismic datasets.
- Apply Transformer-based architectures.
- Perform multi-class seismic event classification.
- Deploy the model for real-time earthquake monitoring.
- Improve robustness against noisy seismic signals.

- **GitHub link**
  https://github.com/FahmidaDipty132/Neural-Networks-Lab/tree/main/NN_project